TPN semantics (5/5)

- An **execution** of a TPN is a path $\omega = s_0 \xrightarrow{\gamma_1} s_1 \xrightarrow{\gamma_2} s_2 \xrightarrow{\gamma_3} \ldots$ such that:
  - $s_0 = \langle m_0, \tau_0 \rangle$ is the initial state
  - γ_i is the i-th fired transition
  - $s_i = \langle m_i, \tau_i \rangle$ is the state reached after the firing of γ_i
  - γ_i is selected among the transitions **firable** in state $s_{i-1} = \langle m_{i-1}, \tau_{i-1} \rangle$
  - after the firing of γ_i, the new marking m_i is derived from m_{i-1} by
    1. removing a token from each input place of γ_i
       (i.e., $m_{\mathrm{tmp}} = m_{i-1}(p) - 1 \; \forall \, p \,|\, (p, \gamma_i) \in A^-$)
    2. adding a token to each output place of γ_i
       (i.e., $m_i = m_{\mathrm{tmp}}(p) + 1 \; \forall \, p \,|\, (\gamma_i, p) \in A^+$)
  - τ_i is derived from τ_{i-1} by:
    1. reducing the time to fire of each persistent transition by the value of $\tau_{i-1}(\gamma_i)$
       (i.e., $\tau_i(t') = \tau_{i-1}(t') - \tau_{i-1}(\gamma_i) \; \forall \, t' \in T \,|\, t'$ is persistent in m_i)
    2. eliminating the time to fire of each transition disabled by the firing of γ_i
    3. sampling the time to fire of each newly enabled transition in its firing interval
       (i.e., $\tau_i(t') \in [EFT(t), LFT(t)] \; \forall \, t' \in T \,|\, t$ is newly enabled in m_i)
  - ω is a finite or infinite path